

S. Vahid Ghayoomie

Scientific Computing & HPC Engineer | Python & C++ | AI & MLOps Engineer | Data Scientist

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Professional Summary

Computational systems engineer with interdisciplinary expertise in high-performance computing, scientific simulation, and AI-driven modeling. Background spans big data architecture, cloud infrastructure, computational neuroscience, and performance-oriented software development. Currently pursuing Master's studies in High Performance Computing with research interests in scalable simulation systems, parallel computing, and emerging quantum machine learning paradigms.

Technical Skills

- **High-Performance & Scientific Computing:** Parallel and distributed computing, numerical methods, large-scale simulation systems, performance-aware software design (Python/C++), HPC workflows, MPI, OpenMP, CUDA
 - **Programming Languages:** Python (NumPy, SciPy, Scikit-learn, TensorFlow, PyTorch, FastAPI, Django), Modern C++ (C++17/20), Bash/Linux scripting; familiarity with MATLAB, R, JavaScript.
 - **AI & MLOps Engineering:** Deep learning, LLM systems (OpenAI, Vertex AI, LangChain), ML pipeline automation (Kubeflow, TFX, MLflow), CI/CD for ML, model monitoring (SHAP, Vertex AI), ONNX, TensorFlow Serving, DVC.
 - **Cloud & Distributed Systems:** GCP (Vertex AI, BigQuery, Dataflow, GKE), Spark/Dataproc, OpenStack, Hadoop ecosystem, AWS & Azure exposure, Docker, Kubernetes, Terraform.
 - **Web Development:** Full-Stack Development (React, Node.js, Django, FastAPI), REST APIs, GraphQL, UI/UX Design (Figma, Photoshop)
 - **Databases & Data Engineering:** PostgreSQL, MySQL, MongoDB, Neo4j, Oracle, SQL Server; large-scale structured and unstructured data systems.
 - **DevOps & Systems Engineering:** CI/CD (GitHub, GitLab), Linux automation, virtualization, high availability, load balancing, container orchestration.
 - **Security & Reliability:** Secure SDLC, OWASP standards, penetration testing, resilient distributed system design
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Professional Experience

Neur Asal Ware (Neursal) | Co-Founder, CEO, IT Specialist | Mar 2022 – Sep 2023

- **MatNeuron:** Designed and developed a **web-based cybernetics simulation environment** integrating neural networks and biomechanical systems.
- **Cloud-Based Software Development:** Led the design and deployment of cloud solutions for e-commerce and real estate platforms, reducing infrastructure costs by **30%**.

- **DevOps Implementation:** Established CI/CD pipelines using **GitLab** and managed cloud environments with **OpenStack**.

Sadad Informatics Corp. - Melli Bank Iran | Big Data Engineer, Data Scientist | Apr 2016 – Sep 2018

- **Big Data Architecture:** Reduced data infrastructure costs by **5x** by implementing **Apache Hadoop, Spark, and OpenStack** on a local cloud cluster.
- **Team Leadership:** Coordinated two teams for big data infrastructure development and maintenance and consulting implementation of a realtime Fraud Detection system

Saba System Sadra Corp. | Software/Hardware Engineer, Project Manager | Mar 2012 – Mar 2015

- **Azaram WAF:** Designed and implemented a **Linux-based web application firewall** with HTTPS offloading, load balancing, and high availability.
- **Security Products:** Contributed to the development of **Mavara SIEM** and **Kandoo Honeybot** for network security monitoring.

Tosan Corp. - Pars Processing Pioneer | Secure Software Development | Jun 2008 – Mar 2012

- **Security Assessments:** Conducted penetration testing and secure code development training for banking and government organizations.
 - **Secure SDLC:** Trained developers on **OWASP, WASC, and Microsoft SDL** guidelines.
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Research Experience

OpenWorm Foundation | Senior Contributor | Nov 2014 – Present

- **ChannelWorm:** Proposed and led the development of a platform for modeling ion channels in **C. elegans**.
 - Built and validated **Hodgkin-Huxley ion-channel models** from experimental patch-clamp datasets.
 - **Developed APIs** for data integration and validation tests

Laboratory of Systems Biology and Bioinformatics, University of Tehran | Research Assistant | Mar 2014 – Sep 2014

- **Ion Channel Structure-Function Analysis:** Developed computational models to predict ion channel kinetics from protein sequences.
 - **Systems Biology of Aging:** Researched aging from a neurobiological perspective.
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Education

- **M.Sc. in High Performance Computing (HPC)**, Politecnico di Milano, Milan, Italy | Sep 2025
- **M.Sc. Studies in Mechatronics Engineering** (Incomplete), Sharif University of Technology International Campus, Kish, Iran | Sep 2019 – Present
 - Relevant Courses: Advanced Mathematical Engineering, Artificial Neural Networks, Advanced Automatic Control

- **M.Sc. Studies in Biology-Biochemistry** (Unfinished Thesis), Payame Noor University, Tehran, Iran | Sep 2011 – 2014
 - Relevant Courses: Genetics, Biochemistry, Biophysics, Advanced Molecular Biology
 - **B.Sc. in Information Technology Engineering**, Islamic Azad University South Tehran Branch, Tehran, Iran | Sep 2005 – Jan 2010 | GPA: 17.43/20 (87%)
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Certifications

- **IBM Data Science Professional Certificate**, Coursera (10 courses completed)
 - **Front-End Web Development with React**, Coursera, March 2019
 - **Practical Machine Learning**, Coursera, June 2015
 - **Python for Genomic Data Science**, Coursera, August 2015
 - **Blockchain for Business**, EDX, May 2018
 - **Introduction to Cloud Infrastructure Technologies**, EDX, July 2018
 - Completed dozens of **Google Cloud Skills Boost** courses, including *Advanced Machine Learning with TensorFlow on Google Cloud*, *MLOps Fundamentals*, *Deploying ML Pipelines on Google Kubernetes Engine*, *BigQuery ML for Data Scientists*, *Building and Deploying Machine Learning Solutions with Vertex AI*, *Responsible AI with Google Cloud*, *Data Engineering*, *Big Data and Machine Learning on Google Cloud*, *Build and Modernize Applications with Gemini*, and *Generative AI with Gemini*.
 - **Credly badges**: <https://www.credly.com/users/vahid-ghayoomie/badges>
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Publications

- **OpenWorm: Overview and Recent Advances in Integrative Biological Simulation of *Caenorhabditis elegans***
GP Sarma, CW Lee, T Portegys, V Ghayoomie, et al. *Phil. Trans. R. Soc. B*, 2018.
 - **Unit Testing, Model Validation, and Biological Simulation**
GP Sarma, TW Jacobs, MD Watts, SV Ghayoomie, et al. *F1000Research*, 2016.
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Conferences

- **Invited Speaker**, The Royal Society Meeting: *Connectome to Behaviour: Modelling C. elegans at Cellular Resolution*, 2018
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Other Activities

- Open Science community contributor.
- Sports enthusiast (swimming, tennis, football, volleyball).
- Multilingual: Fluent in English and Persian; Knowledge of several others
- **IELTS (Feb 2025): Overall 7.0** (Listening: 7.5, Reading: 6.5, Writing: 7.5, Speaking: 6.5).